Hyperspectral Band Selection With Iterative Graph Autoencoder

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***Abstract*— Hyperspectral band selection (BS) is an important task for hyperspectral image (HSI) processing, which aims to select a discriminative and low-redundant band subset. As a significant cue for BS, structure information describes the cross-band correlation, which brings the redundancy of HSI. Existing methods model structure information via manual rule-based graph construction. However, such a graph construction method fails to model complex and diverse structural relationships of HSI data. To address this problem, we propose a data-driven method, named iterative graph autoencoder for BS (IGAEBS). It adaptively captures structure information by a data-specific automatic construction process, rather than by a fixed empirical design. Specifically, we propose a new unsupervised pretext task to train graph convolution neural networks to extract HSI features. These features are used to construct a graph to represent the structural relationships among bands. To enhance the reliability of the graph, we further design an iterative graph improvement mechanism to progressively refine the structure representation. Using the derived graph, we partition the bands into several clusters and select a representative band from each cluster. During the selection process, both intracluster information and intercluster information are considered to improve the discriminativeness of band subset. Extensive experiments are conducted on three public datasets to validate the superiority of the proposed method compared to other state-of-the-art methods.**

***Index Terms*— Graph autoencoder (GAE), graph representation, hyperspectral band selection (BS), representativeness.**

NOMENCLATURE

N	Number of pixels.
B	Number of spectral bands.
t	Number of iterations.
$\boldsymbol{X}$	Spectral matrix.
$\boldsymbol{W}$	Adjacency matrix of band graph.
$\tilde{\boldsymbol{A}}$	Symmetric adjacency matrix.
$\hat{\boldsymbol{W}}$	Prediction adjacency matrix.
$\bar{\boldsymbol{W}}$	Improved adjacency matrix.
$\boldsymbol{Z}$	Feature matrix.
$S^P_{x_i}$	Positive similarity set of the ith band.
$S^N_{x_i}$	Negative similarity set of the ith band.
D_i	Intrasimilarity and intersimilarity distribution distance of the ith band.
I_i	Information entropy of the ith band.
r_i	Ranking value of the ith band.

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I. INTRODUCTION

HYPERSPECTRAL images (HSIs) consist of hundreds of narrow and contiguous spectral bands that contain rich spectral and spatial information [1], [2]. Therefore, HSIs have been applied in many fields, such as agriculture [3], land management [4], [5], geological exploration [6], and military surveillance [7]. However, there are nonnegligible redundant information and noisy bands mixed in original HSIs, which not only lead to the curse of dimensionality [8] but also high computational burden and huge storage cost [9]. Consequently, it is necessary to conduct dimensionality reduction to reduce redundant information and noisy bands [10].

Dimensionality reduction techniques can be divided into feature extraction and band selection (BS) techniques. Feature extraction establishes an appropriate transformation to project data to low-dimensional subspaces; however, although it works well for data dimension reduction, the physical meaning of the new feature space obtained using feature projection is unclear [11]. In comparison, BS methods select a subset of bands from the original data without losing their physical meaning. This approach can provide better interpretability for geological analyses [12].

Existing BS methods can be classified into supervised [13] and unsupervised methods [14]. Considering the difficulty of obtaining sufficient labels, unsupervised BS without using any sample labels is more flexible and applicable than supervised BS. Recently, many unsupervised BS methods have been proposed, which can be divided into four categories: ranking, clustering, searching, and sparsity-based methods. Ranking-based methods aim to evaluate the significance of each band using predefined criteria to choose the top-ranked ones [12], [15], [16]. Clustering-based methods first group bands into several clusters and then select highly uncorrelated bands from the clusters [17], [18], [19], [20]. In searching-based methods, the BS of an HSI is treated as a combinational optimization problem, which can usually be solved using heuristic searching methods [21], [22]. Sparsity-based methods assume that each band can be sparsely represented by several other bands. The

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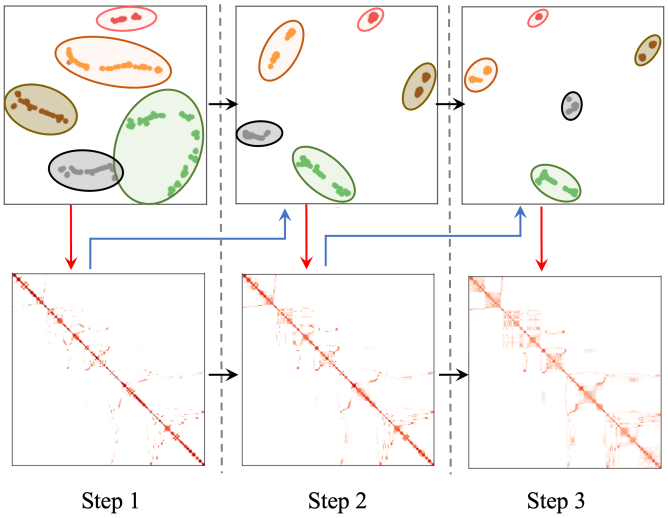


Fig. 1. Data flow of iterative graph improvement mechanism. (Top row) Extracted spectral features in different steps. (Bottom row) Constructed adjacency matrices, which reflect the correlations among bands. Red arrows denote the graph construction process and blue arrows denote the feature extraction process. Black arrows represent the iterative schemes in two processes. With this mechanism, the clustering structure of the extracted features becomes clearer, and more latent relationships among bands are discovered progressively.

estimated sparse coefficients can reveal certain underlying structures within the HSI bands and are used for BS [23], [24]. In addition, some deep learning techniques have been used in HSI BS methods [25], [26] and reinforce learning techniques [27] have been used in HSI BS methods; however, these methods only utilize the statistical properties of HSIs and neglect their intrinsic structural information.

To solve this issue, graph learning-based methods have been introduced into the hyperspectral BS field [28], [29]. In these methods, a graph encoded by an adjacency matrix is constructed to reveal the band correlation. Based on the graph, a selection strategy is then executed to find a proper band subset. The performance of graph learning-based methods is largely determined by the quality of the constructed graph. Thus, the key challenge of BS is to construct a graph that accurately reflects the intrinsic structure of HSI. In [30], the HSI data were assumed to lie in a unified low-dimensional subspace, and the low-rank representation was estimated to construct the graph. Later, the efficient graph convolutional self-representation method [31] extended self-representation into the non-Euclidean domain and used the self-representation coefficient matrix to depict intraband correlations. The hybrid graph convolutional network for BS [32] adopts an elaborately designed feature extraction network to mine the band correlation under a similar self-representation framework. To comprehensively describe the graph, the multigraph integrated embedding and robust self-contained regression method [33] integrated a multigraph from different perspectives. Some methods finely considered spatial information to construct a graph. The multiscale spectral features graph fusion method [34] explored various clustering structures of spectral features at different spatial scales and combined them to obtain a graph. The regionwise multiple graph fusion-based method [35] constructed a regionwise graph for each homogeneous region and designed a diffusion process to fuse the regions to obtain a graph that reflected the global similarity between different bands. Unfortunately, the methods generally use fixed and handcrafted graph construction rules, which limit the capacity in modeling complex and various cross-band structural relationships. Besides, these methods only consider the local structure of HSI in the BS process, while the global structure, i.e., macro-relationship among all of the local structure units, is neglected.

In this article, we propose an unsupervised hyperspectral BS method, named iterative graph autoencoder for BS (IGAEBS), to address these problems. It extracts the structure information through an adaptive data-oriented learning process, rather than a fixed empirical rule. Specifically, we propose a novel unsupervised pretext task to train a graph convolutional neural network (GCN) to extract features of HSI. This task ensures that sufficient structure information is preserved in the features so that we can construct a graph from them to represent the structural relationships among bands. To improve the reliability of the graph, an iterative graph improvement mechanism is further designed to progressively polish the structure representation, as shown in Fig. 1. With the derived graph, we group all bands into clusters accurately, and each cluster describes the local structure of the HSI data. We select one representative band from each cluster to form a band subset. In the selection process, a novel band representativeness metric that calculates the intracluster and intercluster similarity distribution distance (IcSDD) is proposed to make the band subset more discriminative.

The main contributions of this article are summarized as follows.

1) We design an iterative graph autoencoder (IGAE) model to construct a graph that accurately reflects the intrinsic structure of an HSI. Different from the widely used handcrafted graph construction method, this article constructs a graph in a data-driven learnable manner. Specifically, our model can extract features containing abundant structure information while maintaining the original band relationships. Based on the features, we can construct a graph that reflects more latent relationships.
2) We propose a novel metric IcSDD that calculates IcSDD for measuring band representativeness. Compared with existing ranking criteria that only exploits intracluster information, IcSDD can enhance the discriminativeness of the selected bands by using both intracluster information and intercluster information.
3) Comprehensive experiments are performed on three widely adopted hyperspectral datasets. The experimental results demonstrate the superiority of our proposed method over other state-of-the-art hyperspectral BS methods.

II. Preliminary

A. Definition and Notations

We denote a 3-D HSI cube as $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{W\times H\times B}$, where W and H are the width and length of a single band image, respectively, and B is the number of spectral bands. The goal of BS is to seek out a proper band subset with size b ($b < B$)

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such that the redundant bands in the HSI are significantly reduced.

In the following, unless as otherwise specified here, we uniformly use tensors to represent the inputs, outputs, and intermediate outputs involved in the GCNs. In addition, headprint t is used to indicate the tth iteration time. For example, the input adjacency matrix of a GCN in the tth iteration time is denoted as a 2-D tensor $\boldsymbol{W}^{(t)} \in \mathbb{R}^{B\times B}$. The notations of the main variables used in this article are first listed in the Nomenclature.

B. Graph Convolutional Neural Network

The recent development of GCNs has significantly promoted the handling of non-Euclidean data. Most graph convolution operators can be classified into two categories: spectral methods [36], which perform convolution by transforming node representations into the spectral domain using the graph Fourier transform or its extensions, and spatial methods [37], which perform convolution by considering node neighborhoods. In this article, we used a simple but widely used convolution operator, fast approximated convolution, which can be regarded as both a spectral operator and a spatial operator [38]. Considering a graph, $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{A})$, where $\mathcal{V}$ denotes the node set of the graph with $v_i \in \mathcal{V}$, $\mathcal{E}$ indicates the edge set with $\langle v_i, v_j\rangle \in \mathcal{E}$, and $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ represents the adjacency matrix. Formally, if the input feature matrix is $\boldsymbol{X}$ and the adjacency matrix is $\boldsymbol{A}$, then GCN $f(\boldsymbol{X}, \boldsymbol{A})$ performs forward layerwise propagation with the following rule:

$$\boldsymbol{H}_{(l+1)} = \varphi\left(\tilde{\boldsymbol{D}}^{-\frac{1}{2}}\tilde{\boldsymbol{A}}\tilde{\boldsymbol{D}}^{-\frac{1}{2}}\boldsymbol{H}_{(l)}\Theta_{(l+1)}\right). \tag{1}$$

Here, $\tilde{\boldsymbol{A}} = \boldsymbol{A} + \mathbf{I}$ is the adjacency matrix of the undirected graph $\mathcal{G}$ with added self-loops, $\mathbf{I}$ is the identity matrix, $\tilde{\boldsymbol{D}}$ is the degree matrix whose elements are given by $\tilde{\boldsymbol{D}}_{ii} = \sum_j \tilde{\boldsymbol{A}}_{ij}$, and $\Theta_{(l+1)}$ is a layer-specific trainable parameters of the GCN. $\varphi(\cdot)$ denotes a certain activation function, such as the ReLU. $\boldsymbol{H}_{(l)} \in \mathbb{R}^{n\times d}$ is the matrix of activations in the lth layer, where d denotes the dimension of acitvations; $\boldsymbol{H}_{(0)} = \boldsymbol{X}$.

The GCN is essentially built in the non-Euclidean domain; moreover, it has shown extraordinary performance in many traditional problems, including classification [39] and relation inference [40]. Similar to other deep neural networks, GCN can be trained using gradient descent methods.

III. Methodology

In this section, the detailed process of our proposed hyperspectral BS method IGAEBS is provided. Our method consists of three main parts: band graph initialization, iterative graph improvement, and representative BS. A schematic of the proposed method is shown in Fig. 2, and further details are provided in the following.

A. Band Graph Initialization

To consider the spatial and structural information of spectral bands simultaneously, we treat each band as a graph node and the relationship between pairwise bands as a graph edge. First, we reshape the 3-D hyperspectral data to a 2-D spectral matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_B]^{\mathrm{T}} \in \mathbb{R}^{B\times N}(N = W \times H)$, where each spectral feature vector $\boldsymbol{x}_i$ can be viewed as an attribute of a node over the graph.

Then, we construct the adjacency matrix based on a widely used strategy, whose connectivity between nodes is given in [41]. Formally, the connectivity between nodes v_i and v_j is defined as

$$p(v_j|v_i) = \begin{cases} \dfrac{e_{i,k+1} - e_{ij}}{\sum_{m=1}^{k}(e_{i,k+1} - e_{i,m})}, & v_j \in \mathcal{N}_k(v_i) \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where $\mathcal{N}_k$ denotes the k nearest neighbors of v_i in the Euclidean distance, $e_{i,k+1}$ is the $(k+1)$th smallest Euclidean distance between v_i and the other nodes, and e_{ij} is the Euclidean distance between v_i and v_j. Based on the spectral matrix $\boldsymbol{X}$ and initialized sparsity coefficient $\mathcal{K}$, we can compute $p(v_j|v_i)$ by (2). The probability $p(v_j|v_i)$ is specified as the value of the ith row and the jth column of initial adjacency matrix $\boldsymbol{\mathcal{W}}$.

As a result, the initial graph can be represented by a spectral matrix $\boldsymbol{X}$ to denote the spatial information and an adjacency matrix $\boldsymbol{\mathcal{W}}$ to represent the structural information. Note that $p(v_j|v_i) \neq p(v_i|v_j)$ typically holds; therefore, the graph should be viewed as a directed graph. The initial graph only has the limited capacity to represent the structural information of HSI [42]. To learn a more reliable graph, we take it as the initial input of the IGAE.

B. Iterative Graph Improvement

Fig. 2 shows the overall architecture of the IGAE, which consists of two components: a graph autoencoder (GAE) and a graph construction module (GCM). The former was employed to extract the features from a raw HSI, and the latter was employed to construct a graph based on these features. To improve the reliability of the constructed graph, we further design an iterative graph improvement mechanism to progressively refine structure representation. A detailed introduction to the GAE, GCM, and iterative graph improvement mechanism is provided in the following.

1) Graph Autoencoder: GAE consists of an encoder and a decoder. The encoder of GAE is composed of multiple graph convolutional layers and aims to extract the features of HSI. We set the number of graph convolution layers to 2, and the number of nodes in the hidden layer to 128, i.e., the length of the extracted feature vector is 128. Before we transfer the adjacency matrix $\boldsymbol{W}$ constructed by (2) to the encoder, we need to symmetrize it via $\tilde{\boldsymbol{A}} = (\boldsymbol{W} + \boldsymbol{W}^T)/2$. Then, the encoder is employed as a function $f(\boldsymbol{X}, \tilde{\boldsymbol{A}})$ that takes the spectral matrix $\boldsymbol{X}$ and the symmetric adjacency matrix $\tilde{\boldsymbol{A}}$ as inputs, producing a feature matrix of HSI, $\boldsymbol{Z} \in \mathbb{R}^{B\times d}$.

To ensure that the extracted features retain sufficient structure information, we propose a pretext task, reconstructing the band relationships, to guide the encoder. Therefore, we employ the decoder to recover the input adjacency matrix $\boldsymbol{W}$ from the feature matrix $\boldsymbol{Z}$. First, the Euclidean distances between each band feature are calculated by $\hat{e}_{ij} = \|\boldsymbol{z}_i - \boldsymbol{z}_j\|_2^2$. Second, the prediction for the input adjacency matrix is calculated by a SoftMax normalization step; $\hat{\boldsymbol{W}}_{ij} = (\exp(-\hat{e}_{ij}))/(\sum_{j=1}^{B}\exp(-\hat{e}_{ij}))$. Clearly, as $\hat{e}_{ij}$

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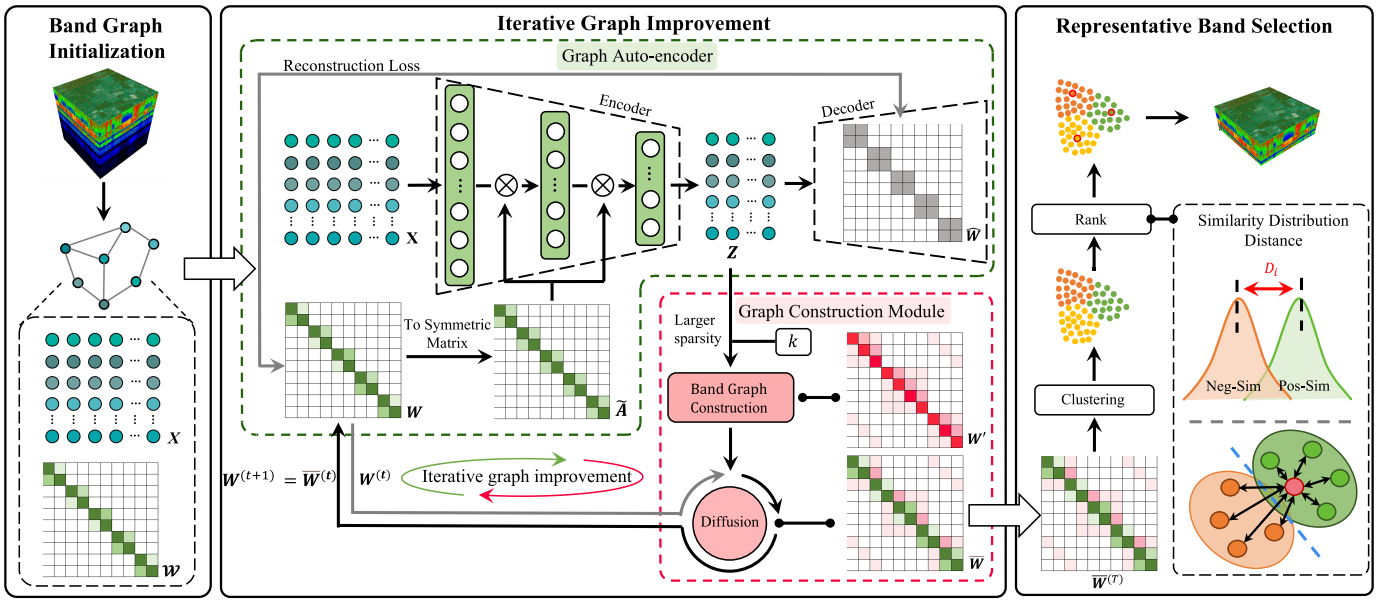


Fig. 2. Architecture of the proposed model. First, an HSI is converted to a graph represented by a spectral matrix and an adjacency matrix. Second, GAE takes the spectral matrix and the initial adjacency matrix as inputs to extract the features of HSI. Then, the GCM utilizes these features to construct a new graph. To improve the reliability of the graph, we use the iterative graph improvement mechanism to refine the representation of HSI structure. In this process, the newly constructed graph is fed as the input to the GAE in the next iteration. After sufficient iterative steps, a reliable graph represented by the refined adjacency matrix is obtained. Finally, based on the graph, all bands are partitioned into several clusters. Also, the representativeness metric IcSDD is applied to rank all bands. The top-rank band in each cluster is selected to form a band subset.

is smaller and $\hat{\boldsymbol{W}}_{ij}$ is larger. In other words, band correlation is measured by the Euclidean distance between band features.

To measure the reconstruction performance, we use Kullback–Leibler divergence as the cost function, denoted as $\mathcal{L}_{\text{rec}}$. Moreover, we desire to keep the connectivity distribution of each node as sparse as possible such that the extracted features present a clear clustering distribution. For this purpose, we use a regularization term $\mathcal{L}_{\text{reg}}$ to constrain the distance between the band features in the loss function

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_{\text{reg}} \\ \mathcal{L}_{\text{rec}} &= \frac{1}{B}\sum_{i=1}^{B}\sum_{i,j=1}^{B} \boldsymbol{W}_{ij} \log \frac{1}{\hat{\boldsymbol{W}}_{ij}} \\ \mathcal{L}_{\text{reg}} &= \sum_{i=1}^{B}\sum_{i,j=1}^{B} \tilde{\boldsymbol{A}}_{ij} \|\boldsymbol{z}_i - \boldsymbol{z}_j\|_2^2 = \text{tr}\left(\boldsymbol{Z}\tilde{\boldsymbol{L}}\boldsymbol{Z}^T\right)\end{aligned} \tag{3}$$

where λ is a regularization coefficient that balances the minimization between the reconstruction error and the regularization term and $\tilde{\boldsymbol{L}} = \tilde{\boldsymbol{D}} - \tilde{\boldsymbol{A}}$. Equation (3) can be optimized using a gradient descent method, such as stochastic gradient descent or adaptive moment estimation.

In GAE, the graph convolution operation is repeated m times so that the features of each band absorb the information of m step neighbors, which depicts the similarity between bands more comprehensively than the raw HSI data. Therefore, based on the features rather than the raw HSI data, we can construct a graph that reflects the intrinsic structure of HSI more accurately.

2) Graph Construction Module: Based on the extracted features, we aim to construct a new graph considering more potential band relationships in GCM. Since graph diffusion can exploit the local manifold information to explore the potential connection between nodes, we design the construction process of new adjacency matrix according to the iterative graph diffusion strategy [43] as follows:

$$\bar{\boldsymbol{W}} = \alpha \boldsymbol{W}' + (1-\alpha)\boldsymbol{W} \tag{4}$$

where $\boldsymbol{W}'$ is the adjacency matrix constructed using (2) based on the feature matrix $\boldsymbol{Z}$, $\boldsymbol{W}$ is the unsymmetrized adjacency matrix of GAE input, and α is a tradeoff parameter that balances the new adjacency matrix and its previous status. In the graph construction process of $\boldsymbol{W}'$, we use a larger sparsity coefficient than the input adjacenct matrix of GAE so that the adjacency matrix $\boldsymbol{W}'$ can represent more potential relationships among bands. By using the tradeoff parameter α, $\bar{\boldsymbol{W}}$ can not only represent new structural relationships but also retain previous structure information.

3) Iterative Graph Improvement Mechanism: We can construct a more reliable graph based on the high-quality features that preserve structure information. Meanwhile, the features can be extracted more accurately when the input graph of the GAE is more reliable. Both processes enhance each other and progressively achieve better performance. Therefore, a reliable graph can be obtained after sufficient iterations. Specifically, for the tth iterative step, where $t = 1, \ldots, T$, GAE is trained to extract features $\boldsymbol{Z}^{(t)}$ under the guidance of adjacency matrix $\boldsymbol{W}^{(t)}$ until the parameters of GAE converge. Following feature extraction, GCM constructs a new adjacency matrix $\bar{\boldsymbol{W}}^{(t)}$ with the features $\boldsymbol{Z}^{(t)}$ and sparsity coefficient $k^{(t)}$. One iterative step is completed with the new adjacency matrix $\bar{\boldsymbol{W}}^{(t)}$, which will guide the feature extraction in the next step as $\boldsymbol{W}^{(t+1)}$. With sufficient steps, we can obtain a reliable graph represented by $\bar{\boldsymbol{W}}^{(T)}$.

In the beginning of the iterative process, the sparsity coefficient is initialized by the predefined constant $\mathcal{K}$, and the input adjacency matrix of GAE is initialized by $\boldsymbol{\mathcal{W}}$. To consider more potential band relationships in each iteration,

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the sparsity coefficient is increased by $\Delta k(\Delta k > 0)$ from the previous step, denoted as $k^{(t+1)} = k^{(t)} + \Delta k$.

During iterative training, our method can exploit the high-quality features of the raw HSI and learn more potential relationships, which helps to correct the wrong relationships and learn new relationships among bands. Therefore, the final graph represented by $\bar{\boldsymbol{W}}^{(T)}$ often reflects the intrinsic structure of the HSI better than the initial graph.

C. Representative BS

Given a certain number of bands b that we want to select, all hyperspectral bands can be partitioned into b clusters by performing a spectral clustering algorithm on the improved graph. Each band cluster can be viewed as a group of redundant bands because they contain similar spectral information. Therefore, we only needed to maintain one band in each cluster.

1) Calculating Intracluster and Intercluster Similarity Distribution Distances for Each Band: Traditional methods usually select the band closest to the centroid or most informative band in each cluster when selecting the representative bands. They treat the clusters independently and ignore valuable information from the intercluster, which is helpful for enhancing the discriminativeness of selected band subset. Hence, we proposed a novel spectral band representativeness metric, IcSDD, which comprehensively exploits intracluster and intercluster information to make the selected band subset distinctive.

We assume that $Y = \{y_i\}_{i=1}^{B}$ denotes the index set of the cluster to which each band belongs. There exists a one-to-one correspondence between y and $\boldsymbol{x}$. Let two bands from the same cluster form a positive pair and two bands from different clusters form a negative pair. We calculate the structural similarity (SSIM) index of a positive pair as $s_{\boldsymbol{x}_i}^P = \langle \boldsymbol{x}_i, \boldsymbol{x}_j \rangle$ s.t. $y_i = y_j$ and the SSIM of a negative pair as $s_{\boldsymbol{x}_i}^N = \langle \boldsymbol{x}_i, \boldsymbol{x}_j \rangle$ s.t. $y_i \neq y_j$. For each band $\boldsymbol{x}_i$, we denote $S_{\boldsymbol{x}_i}^P = \{s_{\boldsymbol{x}_i}^P\}$ and $S_{\boldsymbol{x}_i}^N = \{s_{\boldsymbol{x}_i}^N\}$ as the set of similarities for positive and negative pairs, respectively. The sets of similarities effectively reveal the intracluster and intercluster similarity distributions for each band.

To select a globally representative band in each cluster, we propose a new metric IcSDD, which is the distance between the intracluster similarity distribution and intercluster similarity distribution. IcSDD can describe the representativeness of a band. The representative band of each cluster is always closer to the intracluster bands and more far from the intercluster bands than the nonrepresentative band. In other word, for a representative band, the intracluster similarity tends to be larger, and the intercluster similarity tends to be less. Therefore, the more representative the band, the larger the distance between the intraclass and interclass similarity distributions. We use the Wasserstein metric to calculate IcSDD, i.e., the distance between the intracluster similarity distribution $S_{\boldsymbol{x}_i}^P$ and the intercluster similarity distribution $S_{\boldsymbol{x}_i}^N$

$$\begin{aligned} D_i &= \mathbb{WD}\left(S_{\boldsymbol{x}_i}^P || S_{\boldsymbol{x}_i}^N\right) \\ &= \inf_{\gamma \in \prod(S_{\boldsymbol{x}_i}^P, S_{\boldsymbol{x}_i}^N)} \mathbb{E}_{(s_{\boldsymbol{x}_i}^P, s_{\boldsymbol{x}_i}^N) \sim \gamma} \left[\left\| s_{\boldsymbol{x}_i}^P - s_{\boldsymbol{x}_i}^N \right\| \right] \end{aligned} \tag{5}$$

Algorithm 1 BS Process of IGAEBS

Input: Original HSI $\mathcal{X} \in \mathbb{R}^{W \times H \times B}$, band subset size b, and IGAEBS hyper-parameters.

Initialization: Calculate initial adjacency matrix $\boldsymbol{\mathcal{W}}$ by Eq. (2) with initial sparsity coefficient $\mathcal{K}$.
for $i = 1, 2, \ldots, T$ **do**
  repeat
    Update GAE with Eq. (3) by gradient descent.
  until convergence or exceeding maximum iterations;
  Get new feature $\boldsymbol{Z}^{(t)}$.
  Constrcut $\boldsymbol{W}^{'(t)}$ from $\boldsymbol{Z}^{(t)}$ by Eq. (2) with sparsity $k^{(t)}$.
  Get improved adjacency matrix $\bar{\boldsymbol{W}}^{(t)}$ by Eq. (4).
  Update $\boldsymbol{W}^{(t+1)} = \bar{\boldsymbol{W}}^{(t)}, k^{(t+1)} = k^{(t)} + \Delta k$.
end
Perform spectral clustering on $\bar{\boldsymbol{W}}^{(T)}$.
Calculate intra- and inter- similarity distribution distance D_i for each band by Eq. (5).
Calculate information entropy I_i for each band by Eq. (6).
Compute joint ranking value r_i for each band by Eq. (7).
Select the band with the maximum value of r in each cluster to form the band subset.

Output: Band subset with b selected bands.

where D_i denotes the IcSDD of band $\boldsymbol{x}_i$, $\mathbb{WD}(\cdot)$ denotes the Wasserstein distance, and $\prod(S_{\boldsymbol{x}_i}^P || S_{\boldsymbol{x}_i}^N)$ denotes the set of all joint distributions $\gamma(s_{\boldsymbol{x}_i}, s_{\boldsymbol{y}_i})$, whose marginals are $S_{\boldsymbol{x}_i}^P$ and $S_{\boldsymbol{x}_i}^N$. $\mathbb{E}$ means calculating statistic expectation. inf means finding γ in the joint probability set $\Gamma(S_{\boldsymbol{x}_i}^P, S_{\boldsymbol{x}_i}^N)$ that minimizes the expected value.

2) Selecting Bands: Information entropy [44] is a statistical feature that reflects the average amount of information in the image. To obtain bands that have a large amount of information, we calculate the information entropy (IE) of each band, which is defined as

$$I_i = -\sum_{z \in \omega} p(z) \log p(z) \tag{6}$$

where ω denotes the grayscale space and $p(z)$ is the probability that an event $z \in \omega$ appears in band $\boldsymbol{x}_i$.

From the above analysis, it can be concluded that when both metrics are large, the selected band will not only satisfy a large amount of information but will also be representative. The values of the two metrics do not have the same order of magnitude, and their data must be normalized to the range $[0, 1]$ before being added together. Therefore, the ranking value r_i of each band can be denoted as

$$\begin{aligned} r_i &= \delta(D_i) + \delta(I_i) \\ \delta(l_i) &= \frac{l_i - \min(L)}{\max(L) - \min(L)}, \quad l_i \in L \end{aligned} \tag{7}$$

where l_i is D_i or I_i and $L = \{l_i | i = 1, 2, \ldots, B\}$. According to (7), we select the band with the largest ranking value r_i as the most representative and informative band in each cluster and thus obtain the band subset containing b bands.

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Algorithm 1 summarizes the entire process of the proposed BS method.

In practice, we can determine the number of bands b by the variance-based band-to-power ratio [16], which is defined as

$$R_{\text{var}}(b) = \frac{\sum_{i=1}^{b} \sigma_i^2}{E} \tag{8}$$

where $E = \sum_{i=1}^{B} \sigma_i^2$ is the sum of variances of all the bands. The band-to-power ratio reflects how many bands are needed to produce a certain percentage of band power. Given a desired band-to-power ratio R^*, we can determine the required number of bands b^* by $R_{\text{var}}(b^* - 1) \leq R^* < R_{\text{var}}(b^*)$.

IV. Experiment

In this section, the comparative experiments that were conducted to assess the effectiveness of the proposed BS method are described. First, four commonly used public HSI datasets are introduced in Section IV-A. Second, we introduce the experimental setups involving the comparison methods, evaluation metrics, and parameter settings in Section IV-B. Third, Sections IV-C–IV-F provides the performance comparisons of the proposed method IGAEBS with eight representative unsupervised BS methods performed on four hyperspectral datasets. To further investigate how the performance can be improved by using our method, an ablation study is presented in Section IV-F. Finally, the running time of different methods for selecting a certain number of bands on different datasets is compared in Section IV-G.

A. Datasets

We evaluated our method on four widely used HSI datasets: the Indian Pines, Pavia University, Botswana, and Houston datasets. Information on the four datasets is listed in Table I.

1) Indian Pines: This scene was captured over the Indian Pines test site in Northwestern Indiana and consists of 145 × 145 pixels and 224 spectral reflectance bands in the wavelength range of 0.4–2.5 μm. There are 16 classes of land-cover objects in the scene. Due to the removal of 20 water absorption bands, that is, [104 108], [150 163], and 220, only 200 bands were retained and employed in the experiments.

2) Pavia University: The Pavia University dataset was acquired during a flight campaign over Pavia, northern Italy. The spatial resolution is approximately 1.3 m with a size of 610 × 340, and there are nine classes of land-cover objects in this scene. Similar noisy bands were removed to the Indian Pines dataset, and 103 spectral bands remained.

3) Botswana: The Botswana dataset was gathered by the NASA EO-1 satellite to record a scene from the Okavango Delta, Botswana, in 2001. This image provides a natural scene containing 256 × 1476 pixels for 14 different classes of land cover. Each pixel was represented by 145 useful spectral bands after removing the water absorption and low SNR bands.

4) Houston: The Houston dataset was collected by the Compact Airborne Spectrographic Imager ITRES CASI-1500 in 2012, which covers the University of Houston campus. It includes 349 × 1905 pixels and 144 spectral bands ranging from 346 to 1046 nm. Also, there are 15 classes of land-cover objects in this scene.

TABLE I
Summary of Indian Pines, Pavia University, and Botswana Datasets

Data sets	Indian Pines	Pavia University	Botswana	Houston
Pixels	145×145	610×340	256×1476	349×1905
Channels	200	103	145	144
Classes	16	9	14	15
Labeled pixels	10249	42776	3248	17036
Sensor	AVIRIS	ROSIS	Hyperion	CASI

TABLE II
Hyperparameter Settings for Different BS Methods

Baselines	Indian Pines	Pavia University	Botswana	Houston
MVPCA	-			
E-FDPC	-			
OCF	-			
ASPS	$B = 3$, $M = 10\%$	$B = 10$, $M = 10\%$	$B = 10$, $M = 10\%$	$B = 10$, $M = 10\%$
EGCSR	$\lambda = 1e4$, $k = 5$			
MGSR	$S = 15$, $\lambda = 100$, $p = 0.5$			
BS-Net	$\eta = 2e\text{-}3$, $\lambda = 1e\text{-}2$, $maxiter = 100$			
DARecNet	$\eta = 1e\text{-}2$, $maxiter = 100$			
Ours	$\mathcal{K} = 5$, $\Delta k = 2$, $T = 5$, $\alpha = 0.3$, $\eta = 3e\text{-}4$, $\lambda = 128$, $maxiter = 100$			

B. Experimental Setup

1) Compared Methods: The effectiveness of IGAEBS was evaluated by comparison with eight representative unsupervised BS methods: maximum-variance principal component analysis (MVPCA) [15], enhanced fast density peak-based clustering (E-FDPC) [17], optimal clustering framework (OCF) [18], adaptive subspace partition strategy (ASPS) [19], efficient graph convolutional self-representation (EGCSR) [31], BS network (BS-Net) [25], dual-attention reconstruction network (DARecNet) [26], and marginalized graph self-representation (MGSR) [29]. The hyperparameters settings of all the methods are listed in Table II. Moreover, to comprehensively assess the performance of different methods, using all bands was considered.

2) Evaluation Metrics: We used the classification performance of the selected band subset as its performance indicator, which was widely adopted in the BS evaluation [19], [26]. In our experiments, three typical classifiers, including a support vector machine (SVM) with a radial basis function kernel, k-nearest neighbor (kNN) classifier with $k = 5$, and random forest (RF) with 500 trees, were performed on the selected bands of different BS algorithms. For fairness, in each dataset, 10% of the samples of each class were randomly chosen to construct the training set and the rest for the testing set. Three commonly used quantitative criteria, the overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa), were employed to assess the effectiveness of the proposed methods. In addition, we performed each algorithm ten times and reported the average results to reduce the influence of the random selection of training and testing samples.

3) Parameter Configurations: The hyperparameters are tuned using various approaches. In our experiments, the initial sparsity coefficient $\mathcal{K}$, i.e., the initial neighbor number of a graph node, is set to 5 suggested in [31]. Since the previously learned graph via decoder is taken as the input of encoder in the next iteration, the training process may be unstable if the difference between the inputs of two iterations is large. To avoid this problem, we set the increment Δk in a small value as 2 to ensure that the newly learned graph is not quite

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TABLE III

CLASSIFICATION PERFORMANCE OF BS METHODS USING 13 BANDS ON THE INDIAN PINES DATASET. THE BEST RESULTS ARE SHOWN IN BOLD FONT

No.	MVPCA	E-FDPC	OCF	ASPS	EGCSR	BS-Net	DARecNet	MGSR	Ours IGAEBS
1	42.68 ± 15.40	55.12 ± 14.48	59.27 ± 16.90	63.17 ± 15.17	57.07 ± 10.81	**77.07 ± 9.01**	56.59 ± 10.34	43.17 ± 12.39	58.78 ± 15.13
2	34.01 ± 4.77	48.60 ± 3.82	58.19 ± 1.45	58.48 ± 2.23	58.19 ± 1.91	63.12 ± 2.19	64.62 ± 2.48	43.31 ± 2.77	**64.96 ± 2.65**
3	46.56 ± 3.23	51.58 ± 2.91	57.11 ± 3.61	60.82 ± 3.34	55.78 ± 2.45	57.72 ± 2.49	57.88 ± 2.20	46.53 ± 4.97	**63.32 ± 2.32**
4	35.45 ± 5.94	41.74 ± 2.84	50.33 ± 4.99	56.20 ± 4.96	44.84 ± 6.49	47.32 ± 7.54	52.25 ± 6.90	40.19 ± 5.25	**57.75 ± 6.89**
5	69.84 ± 3.72	79.93 ± 3.43	85.40 ± 2.13	85.38 ± 3.01	81.29 ± 1.65	87.91 ± 3.78	**87.93 ± 2.76**	74.11 ± 3.06	86.02 ± 2.41
6	82.22 ± 4.40	88.37 ± 1.80	87.02 ± 2.26	90.27 ± 2.45	89.71 ± 3.89	94.09 ± 1.44	90.85 ± 2.39	91.51 ± 2.86	**91.95 ± 1.72**
7	37.20 ± 19.84	51.20 ± 20.69	63.60 ± 16.54	74.00 ± 10.47	65.20 ± 13.03	66.40 ± 17.08	71.20 ± 7.96	14.00 ± 7.21	**80.80 ± 8.54**
8	88.63 ± 3.65	95.21 ± 1.97	96.26 ± 1.46	94.86 ± 2.51	94.63 ± 1.43	95.42 ± 2.17	94.58 ± 2.31	82.70 ± 2.81	95.44 ± 2.94
9	20.00 ± 9.03	69.44 ± 15.96	56.67 ± 11.86	**70.00 ± 18.79**	32.22 ± 14.44	52.78 ± 17.61	59.44 ± 18.43	29.44 ± 12.44	**64.44 ± 22.66**
10	50.98 ± 3.22	32.58 ± 3.64	65.98 ± 3.49	67.11 ± 4.08	65.89 ± 3.73	61.63 ± 3.38	68.55 ± 3.94	44.11 ± 3.85	**71.68 ± 3.10**
11	67.82 ± 3.26	70.80 ± 3.10	71.56 ± 1.48	76.21 ± 2.28	73.45 ± 1.92	73.19 ± 2.51	75.99 ± 2.25	61.41 ± 1.63	**76.42 ± 1.66**
12	18.15 ± 3.56	60.15 ± 3.50	69.44 ± 3.86	69.64 ± 5.21	59.72 ± 5.16	57.90 ± 6.97	65.07 ± 3.53	22.42 ± 4.39	**72.58 ± 3.55**
13	73.95 ± 8.66	92.70 ± 3.66	93.73 ± 2.30	94.27 ± 3.25	88.97 ± 4.08	92.38 ± 4.00	95.19 ± 2.98	86.92 ± 4.98	**97.41 ± 1.25**
14	91.67 ± 1.97	92.13 ± 1.57	**93.81 ± 1.31**	92.17 ± 3.09	89.06 ± 2.24	92.18 ± 2.76	90.62 ± 2.34	88.91 ± 1.55	**93.35 ± 2.10**
15	30.40 ± 3.71	44.73 ± 5.59	40.40 ± 3.28	47.84 ± 6.77	39.86 ± 5.74	**54.44 ± 6.03**	43.31 ± 3.85	38.79 ± 4.16	47.12 ± 5.04
16	89.88 ± 3.42	85.12 ± 2.73	90.95 ± 3.62	90.83 ± 2.92	91.31 ± 2.07	90.36 ± 1.55	**91.79 ± 2.79**	87.02 ± 4.92	90.36 ± 3.47
OA	54.96 ± 2.31	66.21 ± 1.69	71.23 ± 2.13	74.45 ± 1.60	67.95 ± 1.80	72.74 ± 2.19	72.87 ± 1.82	55.91 ± 1.21	**75.77 ± 1.92**
AA	59.83 ± 0.66	66.16 ± 0.46	72.38 ± 0.40	74.42 ± 0.72	71.20 ± 0.52	73.31 ± 0.51	74.52 ± 0.35	60.05 ± 0.70	**76.55 ± 0.60**
Kappa	0.54 ± 0.008	0.61 ± 0.011	0.68 ± 0.018	0.71 ± 0.013	0.67 ± 0.009	0.69 ± 0.012	0.71 ± 0.011	0.54 ± 0.013	**0.73 ± 0.008**

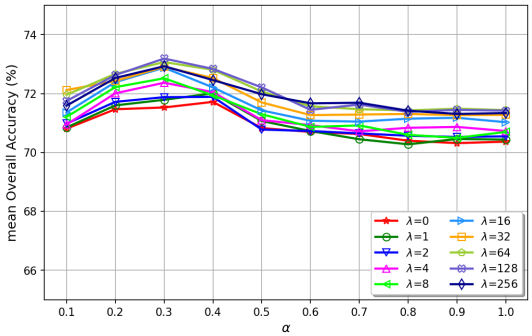


Fig. 3. Influence of parameter variation on BS results on the Indian Pines dataset. The mOA over 3–30 bands at different λ and α values are shown in this figure. The highest mOA is achieved at $\lambda = 128$ and $\alpha = 0.3$.

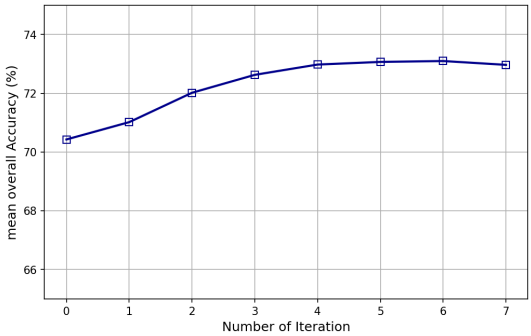


Fig. 4. mOA over 3–30 bands under different iterations on the Indian Pines dataset.

different from the input graph. Meantime, we choose a small learning rate and a relatively large epoch, i.e., η is set to $3e^{-4}$ and maxiter to 100, so that GAE can smoothly converge in each iteration. In addition, we update the input graph with not only the newly learned graph but also the previous input graph to reduce the difference between the inputs of two iterations, which is also helpful to mitigate this problem. As for the tradeoff parameter α and regularization coefficient λ, we vary the value of α from 0.1 to 1 and λ from 0 to 256 to test the mean OA (mOA) over 3–30 bands. The influence of the parameters' variation is shown in Fig. 3. Through comparative trials, we finally choose 128 and 0.3 as the values of λ and α, respectively. To obtain an appropriate value of the number of iterations T, we also vary T from 0 to 7 to test the mOA, which is presented in Fig. 4. As the number of iterations increases, the mOA also gradually improves. However, we find that the performance cannot be further improved by increasing the number of iterations when it reaches 5, so we set T to 5.

C. Results on Indian Pines Dataset

In this experiment, we evaluated the performance of IGAEBS on the Indian Pines dataset. The OA achieved by different BS methods and using all bands under the band subset size ranges 3–30 is shown in Fig. 5(a)–(c). In most cases, the proposed IGAEBS provided the best overall classification results for kNN, SVM, and RF. As shown in Fig. 5(a)–(c), for a band subset size b less than 10, IGAEBS had a performance comparable to that of DARecNet. When b was greater than 10, IGAEBS was superior to the other methods. In particular, when the band subset was small, for example, $b = 13$, IGAEBS achieved a classification performance higher than all bands for both kNN and SVM, whereas the other methods either could not compete with all bands or required more bands.

All the BS methods tended to achieve better classification as the band subset size increased. However, because of the limited labeled samples, the Hughes phenomenon occurred as the band subset size continued to increase. For example, in Fig. 5(a), the OA obtained by ASPS and OCF decreased when the band subset was larger than 15. Despite this, IGAEBS avoided the phenomenon, suggesting that it is capable of comprehensively modeling the intrinsic structure of the spectral bands, resulting in a band subset with less redundancy.

To provide a quantitative comparison with respect to different methods, Table III lists a detailed performance comparison with respect to class accuracy, OA, AA, and Kappa for $b = 13$. From the results in Table III, we can find that IGAEBS outperformed all competitors in terms of OA, AA, and Kappa. Besides, IGAEBS achieved the best class accuracy on 11 ground objects. Correspondingly, we list the wavelengths of 13 bands selected by all compared methods

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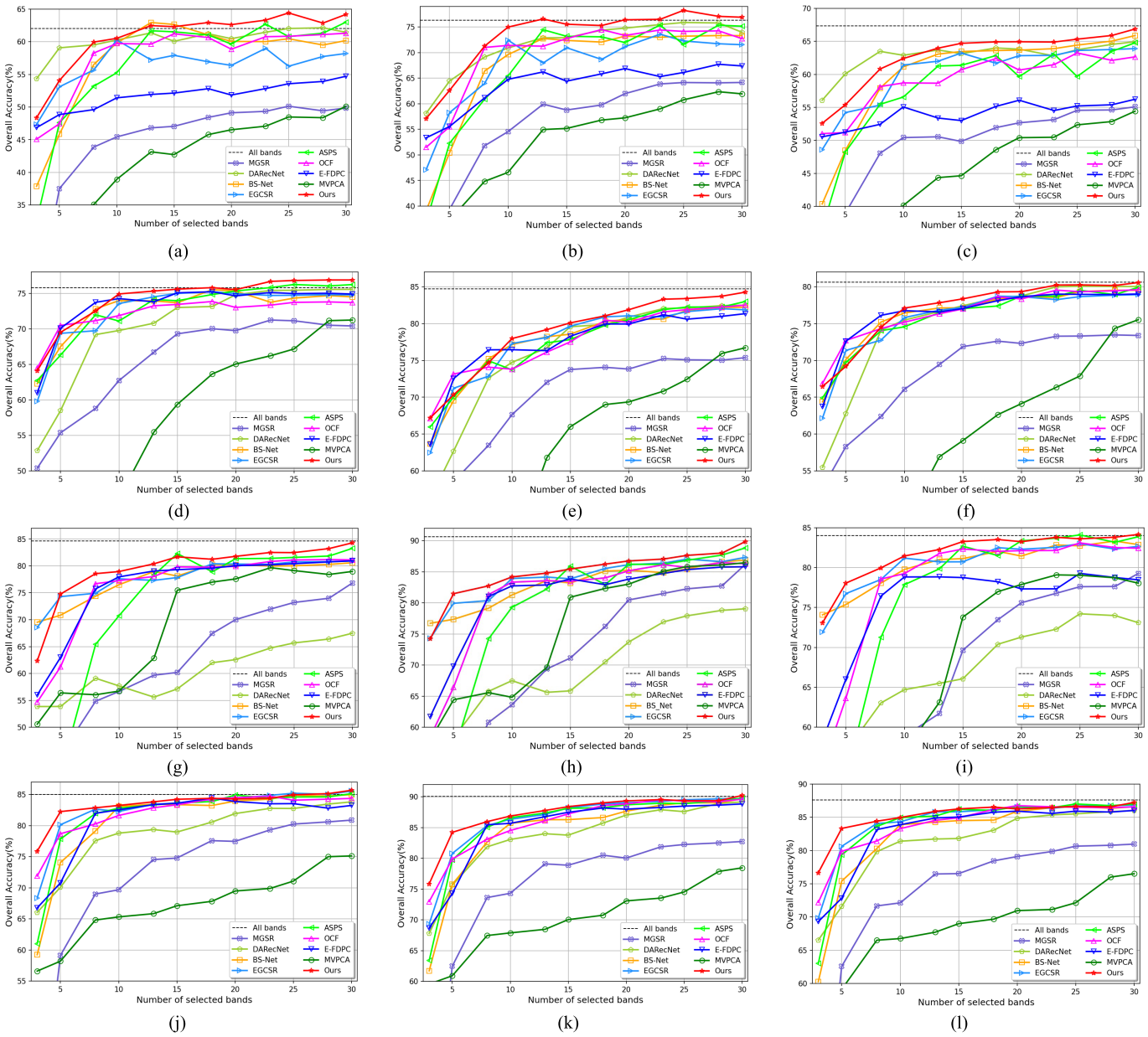


Fig. 5. Classification performance of different BS methods under varying band subset sizes. (a)–(c) Classification results of using kNN, SVM, and RF on the Indian Pines dataset. (d)–(f) Classification results of using kNN, SVM, and RF on the Pavia University dataset. (g)–(i) Classification results of using kNN, SVM, and RF on the Botswana dataset. (j)–(l) Classification results of using kNN, SVM, and RF on the Houston dataset.

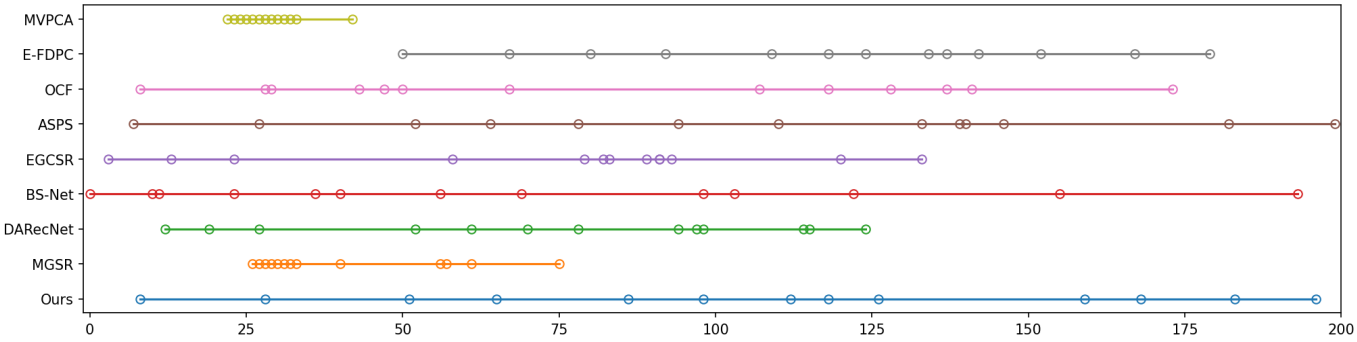


Fig. 6. 13 bands of Indian Pines dataset selected by different methods.

in Fig. 6 and present the classification maps in Fig. 8. As shown in Fig. 6, our selected bands presented a more uniform distribution than others. Based on the fact that the adjacency bands generally include higher correlation, thus, less

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TABLE IV
CLASSIFICATION PERFORMANCES OF DIFFERENT BS METHODS USING 23 BANDS ON THE PAVIA UNIVERSITY DATASET. THE BEST RESULTS ARE SHOWN IN BOLD FONT

No.	MVPCA	E-FDPC	OCF	ASPS	EGCSR	BS-Net	DARecNet	MGSR	Ours IGAEBS
1	77.60 ± 1.15	86.00 ± 1.01	86.93 ± 0.81	86.25 ± 0.91	86.57 ± 0.52	85.58 ± 1.24	87.13 ± 1.03	81.05 ± 1.26	**88.20 ± 1.05**
2	92.83 ± 0.52	92.78 ± 0.40	91.96 ± 0.32	93.26 ± 0.63	93.41 ± 0.40	92.60 ± 0.51	92.40 ± 0.56	**94.11 ± 0.60**	93.51 ± 0.41
3	49.14 ± 2.26	65.94 ± 1.52	65.37 ± 1.46	66.32 ± 1.90	62.12 ± 2.49	62.63 ± 1.26	67.54 ± 2.69	54.72 ± 3.03	**68.27 ± 1.90**
4	77.63 ± 1.54	88.77 ± 1.49	88.74 ± 2.13	89.60 ± 0.98	**90.43 ± 1.04**	89.61 ± 1.18	90.20 ± 1.04	88.84 ± 1.01	89.82 ± 1.63
5	98.98 ± 0.62	99.57 ± 0.30	99.60 ± 0.28	**99.62 ± 0.28**	99.60 ± 0.31	99.60 ± 0.27	99.55 ± 0.36	99.31 ± 0.61	99.60 ± 0.29
6	48.99 ± 1.67	67.79 ± 1.29	70.04 ± 1.53	70.39 ± 1.56	68.92 ± 1.03	68.57 ± 1.76	70.43 ± 1.39	60.49 ± 1.82	**73.63 ± 1.25**
7	35.16 ± 2.73	69.49 ± 2.19	69.41 ± 1.97	70.25 ± 3.06	68.92 ± 2.21	69.01 ± 2.99	69.61 ± 2.84	47.20 ± 3.73	**72.36 ± 1.78**
8	50.28 ± 1.26	66.12 ± 2.32	69.16 ± 2.48	68.18 ± 3.09	64.86 ± 1.37	65.39 ± 2.62	68.74 ± 1.93	58.94 ± 2.22	**70.68 ± 2.70**
9	79.82 ± 2.89	**94.19 ± 0.73**	92.98 ± 1.36	93.13 ± 1.12	93.39 ± 1.03	92.61 ± 1.45	92.77 ± 2.35	92.51 ± 1.48	93.66 ± 1.54
OA	67.83 ± 0.44	81.18 ± 0.32	81.58 ± 0.42	81.89 ± 0.26	80.91 ± 0.35	80.62 ± 0.38	82.04 ± 0.42	75.24 ± 0.72	**83.30 ± 0.41**
AA	76.53 ± 0.31	84.41 ± 0.28	84.66 ± 0.32	85.22 ± 0.33	84.70 ± 0.23	84.15 ± 0.25	85.11 ± 0.39	81.46 ± 0.37	**86.42 ± 0.38**
Kappa	0.68 ± 0.004	0.79 ± 0.005	0.80 ± 0.003	0.80 ± 0.003	0.80 ± 0.004	0.79 ± 0.005	0.80 ± 0.012	0.75 ± 0.011	**0.82 ± 0.007**

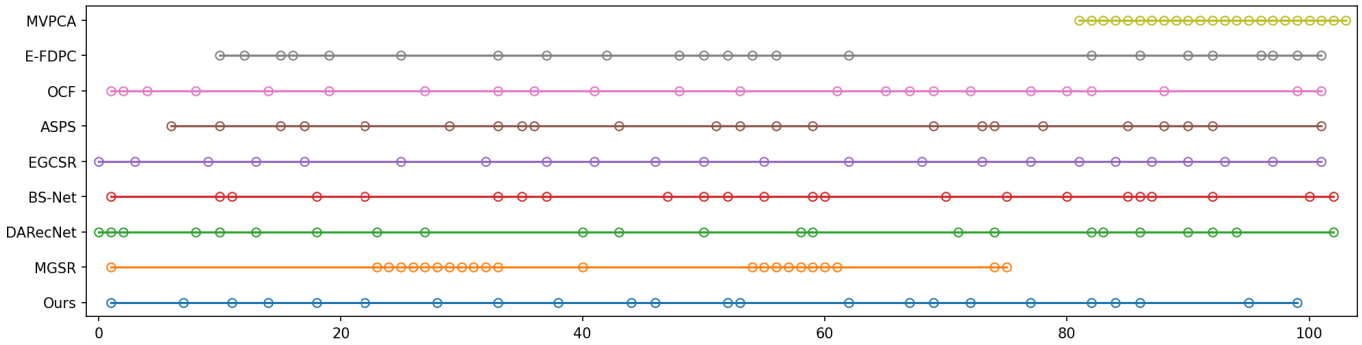


Fig. 7. 23 bands of Pavia University dataset selected by different methods.

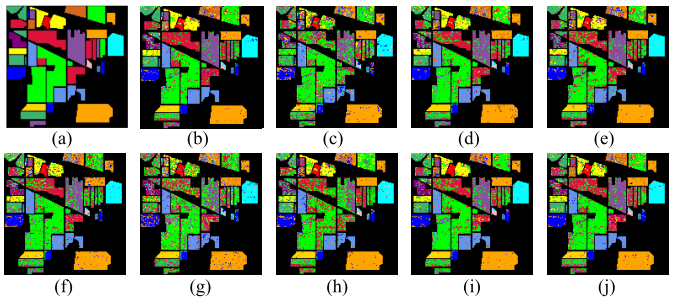


Fig. 8. SVM-based classification maps on the Indian Pines dataset with 13 bands selected using (a) ground truth, (b) Ours IGAEBS, (c) MVPCA, (d) DARecNet, (e) OCF, (f) EGCSR, (g) MGSR, (h) E-FDPC, (i) ASPS, and (j) BS-Net.

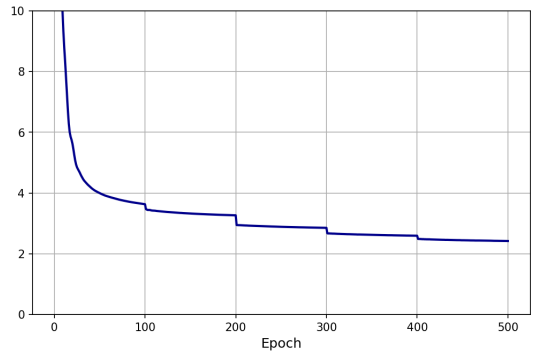


Fig. 9. Loss curve when training on the Indian Pines dataset.

redundancy containing among the band subset was selected by our method. In Fig. 8, our method provided a better classification map, especially in the red and purple regions, with fewer misclassification points than any others, which visually indicates the higher quality of our selected bands.

In addition, we recorded the loss when training on the Indian Pines dataset. As shown in Fig. 9, our model undergoes five iterations, each containing 100 epochs, to reach convergence.

D. Results on Pavia University Dataset

In this section, we present a performance comparison of IGAEBS with eight representative methods on the Pavia University dataset using kNN, SVM, and RF classifiers. Fig. 5(d)–(f) plots the curves of the kNN-, SVM-, and RF-based classification OA against the number of selected bands b under the nine BS methods and all bands. In Fig. 5(d)–(f), we can see that for b less than 10, IGAEBS had comparable performance with E-FDPC, OCF, and EGCSR. When b was between 10 and 20, IGAEBS was marginally superior to the others, and IGAEBS outperformed the other methods for b greater than 20. In Fig. 5(d), IGAEBS achieved a better performance using only 18 bands than all bands, whereas the other methods could not compete with all bands. This reveals that the complete band set contains many noisy bands, which degrade the classification performance. This also suggests that the proposed IGAEBS can neglect noisy bands, resulting in a more informative band subset. A similar tendency can be found in Fig. 5(e) and (f), where IGAEBS used fewer bands to achieve a similar performance with all bands.

Table IV compares the classification performance by setting the band subset size to 23. From Table IV, IGAEBS achieved the best OA, AA, and Kappa values compared to the other methods. In addition, the IGAEBS method was significantly better than the other BS methods in terms of class accuracy.

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TABLE V
CLASSIFICATION PERFORMANCE OF DIFFERENT BS METHODS USING 15 BANDS ON THE BOTSWANA DATASET. THE BEST RESULTS ARE SHOWN IN BOLD FONT

No.	MVPCA	E-FDPC	OCF	ASPS	EGCSR	BS-Net	DARecNet	MGSR	Ours IGAEBS
1	98.85 ± 1.76	99.37 ± 0.76	99.29 ± 0.69	99.01 ± 1.25	99.18 ± 1.12	98.93 ± 1.34	98.77 ± 1.07	96.95 ± 1.70	**99.42 ± 0.81**
2	92.64 ± 4.37	96.26 ± 3.81	93.96 ± 6.39	96.16 ± 2.42	93.08 ± 6.31	**97.69 ± 1.87**	81.43 ± 10.20	88.90 ± 8.50	91.87 ± 6.49
3	83.63 ± 3.26	90.44 ± 3.68	92.08 ± 3.70	90.75 ± 4.50	88.19 ± 3.37	89.20 ± 2.66	80.66 ± 3.99	76.59 ± 3.87	**95.84 ± 1.32**
4	80.67 ± 5.32	90.31 ± 3.00	90.98 ± 3.44	90.93 ± 3.70	92.37 ± 4.67	82.99 ± 5.01	49.54 ± 5.03	67.32 ± 5.21	**92.94 ± 2.62**
5	68.68 ± 6.81	76.24 ± 4.53	77.07 ± 5.79	78.43 ± 3.48	76.94 ± 6.61	**78.88 ± 3.20**	59.34 ± 7.43	70.41 ± 5.08	73.60 ± 6.31
6	57.81 ± 5.76	60.33 ± 5.61	59.75 ± 6.43	67.68 ± 6.04	58.47 ± 8.43	64.71 ± 7.48	36.90 ± 5.71	49.05 ± 6.10	**67.81 ± 9.05**
7	94.33 ± 1.26	97.30 ± 0.77	**97.34 ± 0.83**	96.52 ± 1.14	95.92 ± 1.57	96.35 ± 0.88	96.78 ± 0.80	92.10 ± 3.72	96.91 ± 0.85
8	90.66 ± 3.90	90.00 ± 4.22	89.95 ± 4.08	91.37 ± 2.39	88.47 ± 1.68	91.91 ± 2.07	75.63 ± 4.57	64.92 ± 9.01	**94.10 ± 1.69**
9	**82.33 ± 4.27**	73.85 ± 4.47	66.50 ± 5.40	75.44 ± 3.86	76.50 ± 6.70	73.29 ± 3.62	41.98 ± 5.25	67.46 ± 3.40	81.80 ± 5.21
10	66.23 ± 6.65	72.69 ± 9.12	73.59 ± 6.04	75.61 ± 6.85	73.14 ± 4.95	71.39 ± 5.42	53.18 ± 5.05	40.31 ± 3.03	**79.64 ± 7.97**
11	77.20 ± 5.48	82.36 ± 3.22	81.82 ± 5.34	80.98 ± 5.23	78.58 ± 4.23	80.62 ± 5.83	72.22 ± 3.02	64.00 ± 7.34	**84.25 ± 6.14**
12	76.81 ± 8.22	86.99 ± 4.98	89.08 ± 4.42	**89.88 ± 2.55**	85.83 ± 4.80	84.11 ± 6.06	68.83 ± 7.03	84.91 ± 5.55	81.35 ± 9.00
13	75.89 ± 5.31	70.21 ± 4.30	69.29 ± 4.58	80.09 ± 5.47	78.51 ± 6.53	70.04 ± 5.99	47.18 ± 3.39	64.81 ± 5.61	**80.21 ± 5.46**
14	86.94 ± 4.29	87.41 ± 4.18	87.76 ± 3.98	88.59 ± 3.41	**88.94 ± 1.84**	84.71 ± 4.62	59.53 ± 6.90	68.12 ± 8.26	82.94 ± 3.95
OA	80.91 ± 1.15	83.84 ± 0.65	83.47 ± 1.20	85.82 ± 0.68	83.87 ± 0.97	83.20 ± 1.36	65.85 ± 1.44	71.13 ± 1.16	**85.91 ± 1.10**
AA	79.92 ± 1.15	82.54 ± 0.74	82.02 ± 1.17	84.74 ± 0.86	82.68 ± 0.84	82.02 ± 1.18	65.04 ± 1.25	70.18 ± 0.90	**85.48 ± 0.95**
Kappa	0.78 ± 0.012	0.81 ± 0.012	0.81 ± 0.013	0.83 ± 0.012	0.81 ± 0.012	0.81 ± 0.013	0.62 ± 0.012	0.68 ± 0.013	**0.84 ± 0.012**

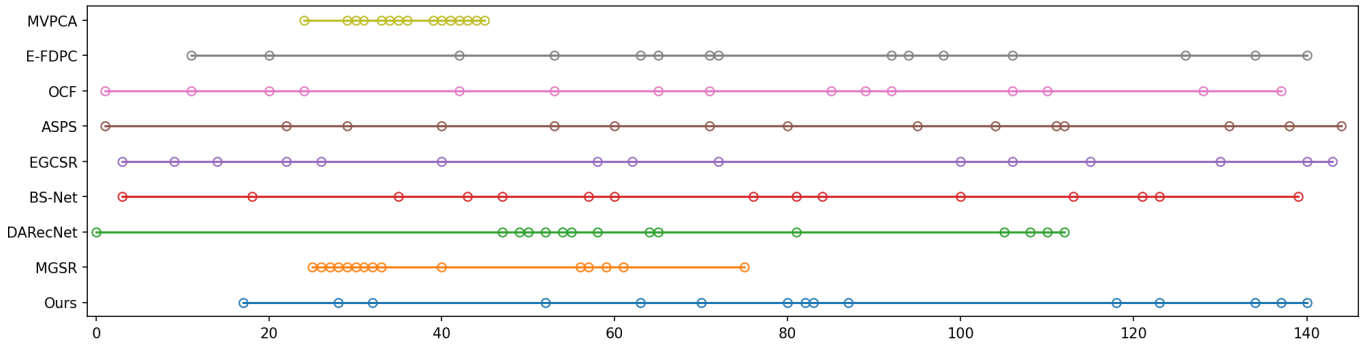


Fig. 10. 15 bands of Botswana dataset selected by different methods.

To better show the band distribution, we indicate the locations of these bands on the spectrum in Fig 7. As we can see, the result obtained by IGAEBS contained less continuous bands with a relatively uniform distribution.

E. Results on Botswana Dataset

We further evaluated the performance of IGAEBS on the Botswana dataset similar to the other two datasets. The performance comparison of the different BS methods on the Botswana dataset is shown in Fig. 5(h)–(j). It can be observed that IGAEBS achieved the best OA under the three classifiers kNN, SVM, and RF. For b greater than 18, the accuracy curves of E-FDPC, OCF, EGCSR, BS-Net, and IGAEBS tended to be stable and very close, whereas for b less than 10, IGAEBS still outperformed most methods. Therefore, the effectiveness of IGAEBS was demonstrated.

The main reason for the robust and excellent performance of IGAEBS is that the IGAE makes the graph consider potentially correlated bands that belong to the same cluster. With the improved graph, reasonable spectral clustering results were obtained. Meanwhile, due to the application of the proposed representativeness metric in band ranking, the discriminative and informative bands can be selected.

Table V lists the classification results obtained using 15 bands and Fig. 10 shows the corresponding band distribution of different methods. IGAEBS was generally superior to the other BS methods in most classes and significantly outperformed all the other BS methods in terms of OA, AA, and Kappa. In addition, the band subset selected by IGAEBS contained fewer continuous bands than MVPCA, MGSR, and DARecNet, showing that the selected band subset contains less redundant information.

F. Results on Houston Dataset

We evaluated the performance of IGAEBS on the Houston dataset and compared its performance to other BS methods using three classifiers: kNN, SVM, and RF. The results are presented in Fig. 5(j)–(l), where IGAEBS achieved superior OA compared to other methods. For b greater than 20, the accuracy curves of ASPS, OCF, EGCSR, and IGAEBS tended to be stable and very close, whereas for b less than 20, IGAEBS outperformed most methods.

Table VI lists the classification results obtained using 20 bands and Fig. 11 shows the corresponding band distribution of different methods. IGAEBS was generally superior to the other BS methods in most classes and outperformed all the other BS methods in terms of OA, AA, and Kappa. In addition, the band subset selected by IGAEBS contained fewer continuous bands than other methods, such as MGSR, BS-Net, and DARecNet. This suggests that the selected band subset contains less redundant information.

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TABLE VI

CLASSIFICATION PERFORMANCE OF DIFFERENT BS METHODS USING 20 BANDS ON THE HOUSTON DATASET. THE BEST RESULTS ARE SHOWN IN BOLD FONT

No.	MVPCA	E-FDPC	OCF	ASPS	EGCSR	BS-Net	DARecNet	MGSR	Ours IGAEBS
1	83.51 ± 2.12	95.80 ± 1.27	96.69 ± 0.43	96.18 ± 1.33	96.77 ± 0.87	96.26 ± 1.19	96.61 ± 1.14	92.88 ± 2.03	**96.84 ± 1.22**
2	95.84 ± 1.43	96.58 ± 1.28	96.97 ± 1.11	97.36 ± 1.20	96.73 ± 1.06	97.11 ± 1.20	96.42 ± 1.45	96.56 ± 1.20	**97.27 ± 1.00**
3	98.77 ± 0.49	99.10 ± 0.23	99.20 ± 0.26	99.19 ± 0.28	99.16 ± 0.27	99.20 ± 0.27	99.19 ± 0.25	99.47 ± 0.36	**99.32 ± 0.28**
4	80.96 ± 2.22	97.35 ± 0.91	98.30 ± 0.69	97.48 ± 0.96	98.23 ± 0.91	97.27 ± 1.00	97.55 ± 0.89	92.97 ± 2.28	**98.46 ± 0.87**
5	96.50 ± 0.82	98.95 ± 0.33	98.97 ± 0.48	98.72 ± 0.53	98.81 ± 0.52	98.87 ± 0.36	98.71 ± 0.49	90.86 ± 1.44	**98.92 ± 0.54**
6	**98.49 ± 1.05**	96.07 ± 2.21	97.05 ± 1.72	96.66 ± 1.65	96.79 ± 1.61	96.95 ± 1.50	96.03 ± 2.84	92.89 ± 2.74	96.95 ± 1.64
7	59.74 ± 1.50	90.98 ± 1.64	91.35 ± 1.33	90.45 ± 2.13	**92.65 ± 2.07**	91.07 ± 1.90	86.95 ± 1.62	81.27 ± 2.65	90.87 ± 1.35
8	53.50 ± 2.41	86.16 ± 1.38	87.17 ± 1.72	86.27 ± 1.46	87.31 ± 1.46	85.17 ± 1.44	86.82 ± 1.77	73.59 ± 1.36	**87.63 ± 0.95**
9	63.91 ± 2.38	78.92 ± 3.05	80.44 ± 2.96	80.48 ± 3.07	**81.68 ± 3.23**	79.45 ± 2.94	75.88 ± 2.01	67.92 ± 2.11	79.81 ± 2.16
10	54.27 ± 3.49	84.89 ± 2.20	85.04 ± 1.78	85.21 ± 1.34	85.83 ± 1.92	81.96 ± 1.03	81.55 ± 1.82	59.96 ± 2.44	**85.93 ± 2.16**
11	53.63 ± 2.40	77.76 ± 2.58	78.59 ± 0.46	**78.89 ± 1.36**	81.70 ± 1.33	78.01 ± 1.68	71.33 ± 2.16	71.88 ± 2.91	78.34 ± 1.80
12	36.08 ± 1.59	73.27 ± 2.30	74.88 ± 3.08	74.09 ± 2.46	74.71 ± 1.51	69.22 ± 2.57	73.99 ± 3.85	45.45 ± 2.04	**75.47 ± 1.96**
13	29.07 ± 2.87	48.75 ± 4.51	52.28 ± 3.89	53.81 ± 4.36	53.08 ± 2.19	49.56 ± 2.37	48.73 ± 4.39	44.59 ± 2.83	**55.17 ± 4.48**
14	94.98 ± 2.07	97.55 ± 1.13	97.88 ± 1.25	97.77 ± 1.27	97.19 ± 1.24	97.21 ± 1.26	97.64 ± 1.31	93.57 ± 2.48	**97.94 ± 1.12**
15	97.01 ± 0.97	97.73 ± 0.93	97.91 ± 0.80	97.95 ± 0.92	98.05 ± 0.87	97.80 ± 0.96	97.98 ± 0.80	96.74 ± 0.68	**98.40 ± 0.88**
OA	73.08 ± 0.44	87.99 ± 0.46	88.85 ± 0.49	88.70 ± 0.45	89.05 ± 0.34	87.67 ± 0.19	87.03 ± 0.35	80.04 ± 0.54	**89.34 ± 0.44**
AA	70.11 ± 0.43	87.71 ± 0.52	88.52 ± 0.46	88.31 ± 0.48	88.86 ± 0.43	87.24 ± 0.33	86.40 ± 0.49	78.48 ± 0.44	**89.03 ± 0.45**
Kappa	0.68 ± 0.011	0.87 ± 0.012	0.88 ± 0.013	0.87 ± 0.012	0.88 ± 0.012	0.86 ± 0.011	0.85 ± 0.013	0.77 ± 0.012	**0.89 ± 0.013**

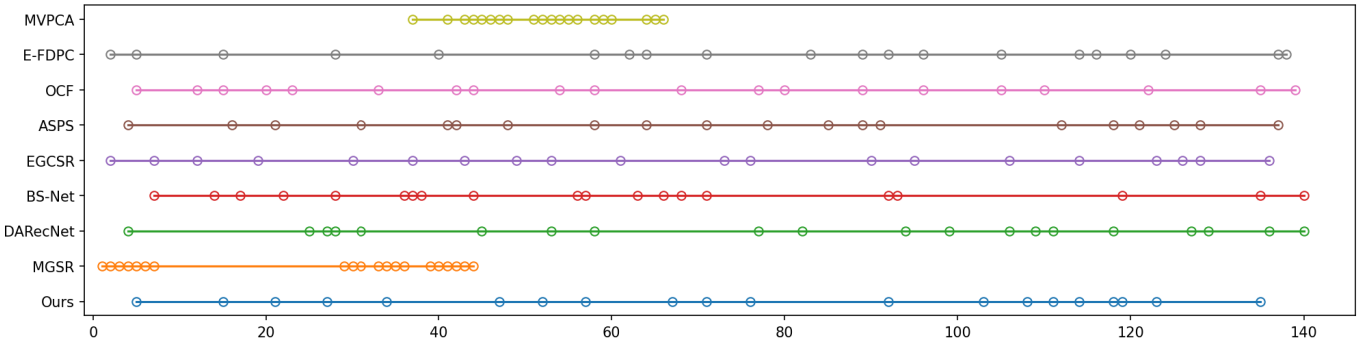


Fig. 11. 20 bands of Houston dataset selected by different methods.

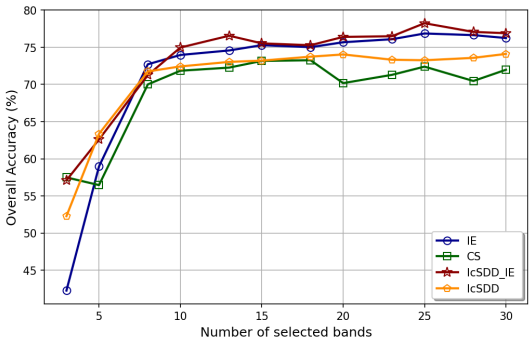


Fig. 12. SVM-based classification performance of different evaluation metrics under varying band subset sizes.

TABLE VII

ABLATION ANALYSIS OF THREE DATASETS

Mehtod	Indian Pines mOA	Pavia University mOA	Botswana mOA
w/o IGAE-w/o IcSDD	69.80	76.36	82.46
w/o IGAE-w/ IcSDD	70.56	77.83	83.70
w/ IGAE-w/o IcSDD	71.82	78.02	84.27
w/ IGAE-w/ IcSDD	**73.19**	**78.91**	**84.74**

G. Ablation Study

We further implemented an ablation study to measure the effectiveness of the IGAE and the metric IcSDD. Table VII presents the classification performance of the methods with and without IGAE and IcSDD on the three datasets using the SVM classifier. mOA is the mean overall accuracy over 3–30 bands. w/ IGAE and w/o IGAE are methods with or without IGAE, respectively. w/ IcSDD and w/o IcSDD are methods with or without IcSDD, respectively. The best results are given in bold. As can be seen, IGAE and IcSDD effectively improved the BS performance. We conjecture that there are two reasons for this. First, IGAE adaptively constructs a graph to represent the structure information of HSI by a data-specific way. The derived graph can accurately reflect the intrinsic structure of HSI, which effectively promotes band clustering. Second, in representative BS, the proposed metric makes full use of the intracluster and intercluster band distribution information to comprehensively measure band representativeness. Intracluster information can reduce the redundancy of band subsets, and intercluster information can improve band representativeness. Therefore, the ranking criterion based on it is helpful for selecting discriminative bands.

To validate the effectiveness of different evaluation metrics, we test the BS performance of different evaluation metrics under varying band subset sizes on the Indian Pines dataset. As shown in Fig 12, combining IcSDD and IE (IcSDD_IE) is generally better than using only IE, IcSDD as an evaluation metric. In addition, our method outperforms

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TABLE VIII
COMPUTATION TIME (S) FOR SELECTING 20 BANDS USING DIFFERENT METHODS

Method	Indian Pines	Pavia University	Botswana
MVPCA	0.16	1.09	3.12
E-FDPC	0.17	0.68	1.83
OCF	0.62	0.99	2.55
ASPS	1.22	2.37	3.05
EGCSR	9.43	15.39	6.26
MGSR	13.26	28.17	9.38
DARecNet	233.84	496.47	212.02
BS-Net	288.59	533.97	262.81
Ours IGAEBS	164.54	309.70	128.77

TABLE IX
RUNNING TIME OF DIFFERENT ITERATION NUMBERS ON THE INDIAN PINES DATASET

Iteration	1	2	3	4	5	Total
Time(s)	20.06	20.65	20.81	20.80	20.85	103.17

the centroid-based selection strategy (CS) even when only using IcSDD. This is probably because our method uses both intracluster information and intercluster information to select representative bands, and meanwhile, CS exploits only intracluster information.

H. Comparison of Running Time

To analyze the running time, we conducted the BS methods on the same computer and recorded their running time. MVPCA, OCF, and E-FDPC were implemented in MATLAB, while the other methods were implemented in Python. We trained IGAEBS to construct a band graph on the GPU platform and calculate the representativeness metric IcSDD on the CPU because the GPU is more friendly for model training. The results are presented in Table VIII. The running time of our method was significantly shorter than that of the other deep learning-based methods, BS-Net and DARecNet. Because IGAEBS required substantial iterative training to construct a band graph, it showed a much greater computational burden than the algebraic methods, MVPCA, E-FDPC, OCF, ASPS, EGCSR, and MGSR. More importantly, the proposed method achieved the best classification performance. In summary, the proposed IGAEBS balances the classification performance and running time.

We further tested the time of the iterative graph improvement. As shown in Table IX, five iterations of the iterative graph improvement take a total of 103.17 s and each iteration takes about 20.80 s. Since we just need to train a small-scale GAE, the convergence time with multiple iterations is not huge.

V. CONCLUSION

This article presents an unsupervised BS method IGAEBS, which can adaptively capture structure information of HSI to promote BS. The proposed method utilizes a novel pretext task that aims to reconstruct the learned band relationships to guide convolutional networks to discover the latent structure of the HSI data. Relying on the task, the IGAE is employed to construct a graph that accurately reflects the intrinsic structure of HSI. To further improve the reliability of the constructed graph, we design an iterative graph improvement mechanism to train the model to refine the structure representation. Using the reliable graph, all hyperspectral bands are partitioned into a few reasonable clusters using the spectral clustering algorithm. Then, one band in each cluster is selected to form a band subset. During the selection process, the representative metric IcSDD is adopted to rank bands, which improves the disciminativenss of the selected band subset. Comparative experiments with several classical BS methods are conducted on Indian Pines, Pavia University, Botswana, and Houston datasets. Through the experiments, we can find that IGAEBS constantly performs well and is robust on different datasets.

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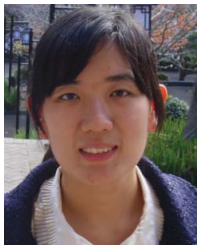
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